OFFICIAL PROJECT DOCUMENTATION | MICROSOFT POWER BI & DAX ANALYTICS

Project Documentation — Zillow Home Value Index (ZHVI)

End-to-End Business Intelligence Documentation: Team Governance, Business Questions, KPI Dictionary, Power Query ETL, Star-Schema Data Modeling, Table View Calendar Engineering, DAX Measures, and Interactive Dashboards

Project Executive Summary & Objective: This business intelligence project in **Microsoft Power BI** analyzes the pricing behavior and growth trajectories of U.S. residential real estate across Metropolitan Statistical Areas (MSAs) from **January 2000 through 2026** (with dynamic filtering for horizons such as 2020–2026). Built on the official **Zillow Home Value Index (ZHVI)** dataset, the system transforms wide, unformatted CSV records into an interactive decision-support platform that identifies high-growth markets, reasonably priced entry regions, and long-term compounding stability.

Project Workflow & 20-Page Documentation Architecture

GOVERNANCE & STRATEGY (PAGES 2–3)

Team Roles, Business Questions & KPIs

Complete breakdown of project team responsibilities across all 5 phases, executed business questions, and the implemented KPI dictionary categorized into historical return, entry cost, and momentum.

PHASE I & II: DATA PREPARATION (PAGES 4–6)

Raw Excel Diagnostics & Power Query ETL

Inspection of raw CSV structural defects in Excel followed by 8 sequential Power Query M transformations (Unpivot, typing, and filtering) achieving 100% valid data quality.

PHASE III: ANALYTICS ENGINE (PAGES 7–14)

Model View, Table View & 10 DAX Measures

Corrected Star-Schema Model View (Page 7), Calendar Table View inspection (Page 8), Measure Container architecture (Page 9), and full DAX code for all 10 measures including the live CAGR % formula capture.

PHASE IV & V: DELIVERY & QA (PAGES 15–20)

Visual Dashboards & System Architecture

Visual walkthrough of all 4 interactive dashboards (Home, Overview, Investment Insights, Geographic Analysis), internal PBIX binary architecture, and QA maintenance guides.

Documentation Module	Pages	Core Technical Deliverables & Verified Visual Captures
Team Governance & KPI Strategy	Pages 02 – 03	5-Phase Team Roster (including **Alaa Mohamed Mahmoud Genidy**), 5 Business Questions, 7-Row KPI Dictionary.
Raw CSV Audit & Power Query ETL	Pages 04 – 06	Raw Excel CSV view (Figure 1), Power Query 100% Valid profile (Figure 2), 8 Applied M-Code Steps.
Star Schema, Table View & DAX Architecture	Pages 07 – 09	Corrected Model View diagram (Figure 3), Calendar Table View (Figure 4 — 9,710 rows), 10-Measure Dependency Matrix.
DAX Formula Library & Live CAGR % View	Pages 10 – 14	Full DAX syntax for Measures 1–10 + Live Power BI Formula Bar capture of CAGR % (Figure 5).
Interactive Dashboards & PBIX Package	Pages 15 – 20	Home, Overview, Investment Insights, Geographic screens (Figures 6–9) & File Explorer structure (Figure 10).

2000–2026
Full Dynamic Horizon

100% Valid
Post-ETL Data Quality

10 Measures
Implemented DAX KPIs

4 Dashboards
Interactive Report Pages

Project Team Structure, Roles & Engineering Deliverables

Team Governance

Collaborative Engineering Workflow: The Zillow Home Value Index (ZHVI) project was executed across five structured engineering phases, progressing from business requirements and raw CSV cleaning to Star-Schema modeling, interactive UI/UX development, and comprehensive quality assurance.

Phase	Team Member Name	Role & Domain	Key Responsibilities & Deliverables
Part 1 Planning	**Amira Mohamed Gomaa Ali**	Business Analysis & Dashboard Design	• Analyzed dataset structure and column semantics. • Defined Business Questions, KPIs, and mathematical formulas. • Selected dashboard pages, visual mappings, and initial wireframes. **Deliverables:** *KPI Dictionary + Business Questions + Wireframe.*
Part 2 Data Prep	**Aya Mostafa Ahmed AbuBakr**	Data Cleaning & Power Query ETL	• Audited raw data quality, missing values, and duplicates in CSV. • Filtered target MSA rows and excluded country-level aggregates. • Executed `Unpivot` converting Wide to Long format and standardized types. **Deliverables:** *Clean Dataset + Power Query Steps + Quality Notes.*
Part 3 Analytics	**Mariam Ahmed Abdelmuttalib**	Data Modeling & DAX Engineering	• Designed the Star-Schema Data Model and `1:*` active relationships. • Built the dedicated Calendar `Date` table (9,710 rows) and `Start Year Selector`. • Authored and validated all 10 analytical DAX measures inside `Dax Measures`. **Deliverables:** *Data Model + DAX Measures + Validation Results.*
Part 4 Visualization	**Alaa Mohamed Mahmoud Genidy**	Dashboard UI/UX Development	• Implemented the approved wireframe design inside Power BI Desktop. • Distributed KPI cards and analytical charts across 4 interactive pages. • Configured interactive slicers, cross-filtering, and navigation buttons. • Unified corporate Zillow color palettes, typography, and visual formatting. **Deliverables:** *Fully Integrated & Interactive Power BI Dashboards.*
Part 5 Delivery & QA	**Ahmed Shabaan Abdelhamed**	QA, Publishing & Documentation	• Tested all KPI cards, DAX measures, and interactive slicers across horizons. • Cross-validated calculations with Part 3 and UI layout with Part 4. • Prepared final PBIX package, GitHub `README.md`, and 20-page PDF documentation. **Deliverables:** *Final Tested PBIX + Technical Documentation + Presentation.*

5 Phases	5 Specialists	100% QA	GitHub Ready
End-to-End BI Lifecycle	Cross-Functional Team	Cross-Validated Metrics	Full Repository Docs

Executed Business Questions & Implemented KPI Dictionary

Strategy & KPIs

Core Business Question Answered	Answering Dashboard & Visual	Actual Project Finding (2000–2026 Baseline)
1. What is the average home value and overall growth across markets during the period?	*Overview* & *Investment Insights* KPI Cards + Trend Area Chart	Period Average: **$192.79K** | Latest Value: **$300K** | Cumulative Growth: **169.82%** | Compound Annual Growth (CAGR): **3.80%**.
2. Which markets achieved the highest and lowest growth over the selected horizon?	*Overview* Top/Lowest KPI Cards & Top 10 / Bottom 10 Bar Charts	**Highest Metro:** `Kapaa, HI` (**552%** total / **7.31%** CAGR), followed by `Prineville, OR` (**414%**). **Lowest Metro:** `Camden, AR` (**42%**) & `Forrest City, AR` (**42%**).
3. Which markets combine strong historical growth with reasonable entry prices?	*Investment Insights* Scatter Plot (Growth vs. Latest Home Value)	Metros in the upper-left/mid quadrant (`Prineville, OR` and `Steamboat Springs, CO`) deliver **>400%** growth without extreme $1M+ entry costs.
4. Is market growth continuing in the recent year, or is momentum slowing down?	*Overview* & *Investment Insights* `YoY Growth %` KPI Card	Trailing 12-month growth stands at **+2.26% YoY**, confirming positive appreciation continues, though moderating vs. the **3.80%** long-term CAGR.
5. How do individual states compare, and how does any market differ from the benchmark?	*Geographic Analysis* Choropleth Map, State Bar Chart & Slicers	**Hawaii** leads all states (>400% state average growth), followed by **New Hampshire** and **Maine**; slicers benchmark any market dynamically.

Implemented Investment KPI Dictionary (Categorized by Analytical Purpose)

Implemented KPI	KPI Category	Calculation Method in DAX	Strategic Value to Investor
Cumulative Growth %	Historical Return	Relative change between Home `Value (Start)` and `Latest Home Value`	Measures total capital appreciation generated across the holding period.
CAGR %	Return & Stability	Compound Annual Growth Rate using exact day fraction (`365.2425`)	Standardizes annual compounding speed and ranks the **Top 5 Most Stable Regions**.
Latest Home Value	Entry Cost	Average property value at final recorded date (**$300K**)	Estimates current capital required to enter a regional market.
Home Value (Average)	Entry Baseline	Multi-year mean property value across selected horizon (**$192.79K**)	Provides historical price baseline against which current valuation is compared.
YoY Growth %	Recent Momentum	Trailing 12-month change via `EDATE(CurrentDate, -12)` (**2.26%**)	Reveals whether price appreciation is accelerating or cooling in the latest year.
Highest / Lowest Region	Market Extremes	Dynamic `TOPN(1, ...)` ranking across all unmasked metro areas	Automatically surfaces #1 outperformer (`Kapaa, HI`) and #1 laggard (`Camden, AR`).
State Average Growth %	Spatial Benchmark	Unweighted AVERAGEX across unique `RegionIDs` per state	Prevents expensive megacities from distorting state-level geographic comparisons.

Methodological Scope Note: ZHVI measures residential home value appreciation and does not include rental yields, property taxes, or financing costs. These KPIs serve as a macro market-screening mechanism to shortlist high-potential MSAs for deeper due diligence.

Phase I: Raw Data Diagnostics in Excel / CSV Source

Data Audit

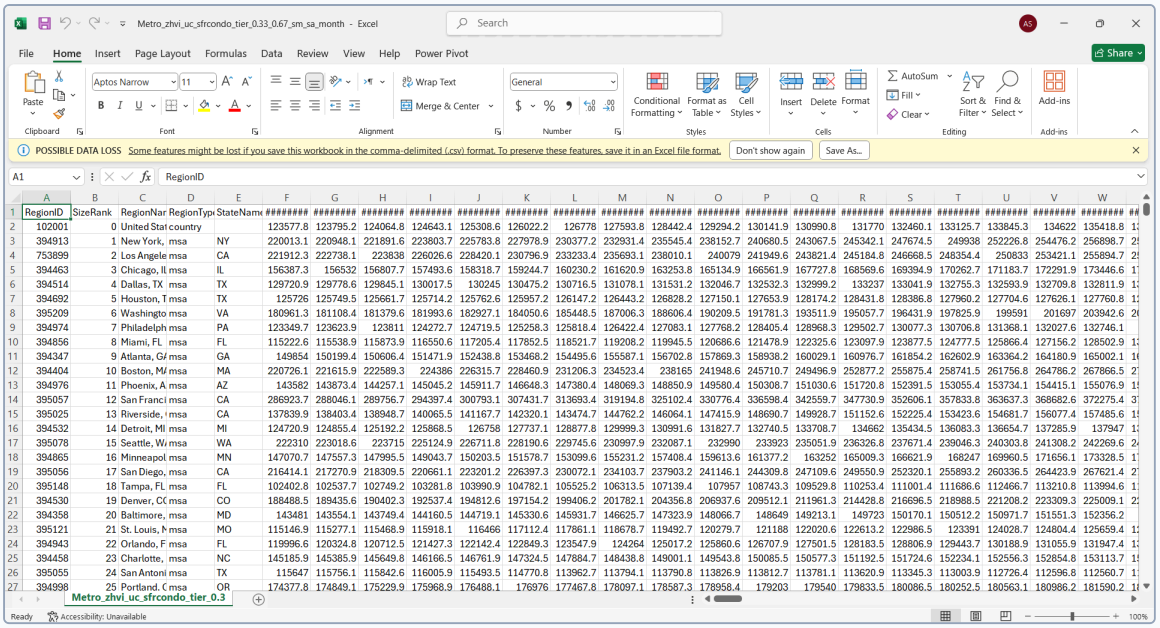


Figure 1: Inspection of raw source file (Metro_zhvi_uc_sfrcondo_tier_0.33_0.67_sm_sa_month.csv) in Excel prior to ETL cleaning

Identified Source Issue	Excel Visual Manifestation	Analytical Impediment in Power BI
1. Wide Date Layout Wide Table Format	Columns F through W+ represent sequential months and years horizontally across hundreds of columns.	Each month residing in a separate column prevents Power BI from constructing continuous time-series curves or establishing relationships with a Calendar dimension.
2. Formatting Corruptions Display Errors	Display of ######## overflow tokens in Row 1 date headers and unformatted decimal strings.	Dates are stored as wide text headers lacking proper `Date` data typing, while identifier columns lack explicit integer constraints.
3. National Aggregate Row United States Row	Row 2 (`RegionID: 102001`, `SizeRank: 0`) holds `RegionName = United States` and `RegionType = country`.	Mixing a national summary row alongside metropolitan city records (`RegionType = msa`) contaminates regional rankings and distorts aggregate calculations.
4. Missing State Values Blank StateName	The `StateName` cell in Row 2 (E2) is completely blank/empty.	Unassigned state cells generate a `(Blank)` category inside state slicers and choropleth maps, compromising geographic integrity.

Phase II (A): Power Query ETL — Source Import & Header Promotion

ETL Steps 1–4

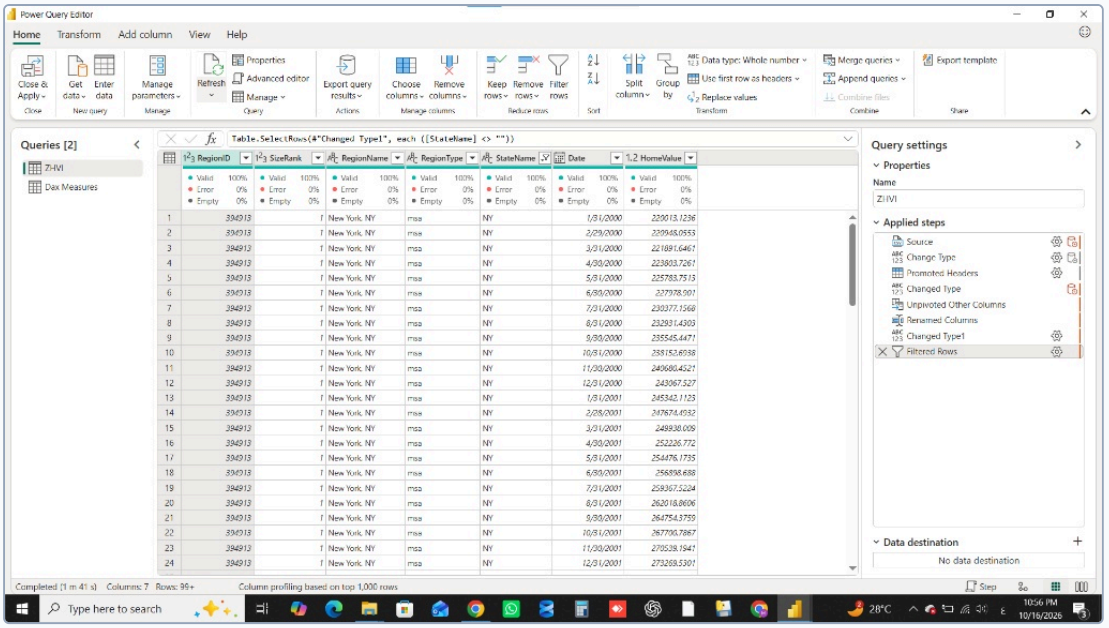


Figure 2: Power Query Editor interface after cleaning the ZHVI table, confirming 100% Valid data quality across all 7 columns

Applied Step Identifier	M-Code / Transformation Function	Technical Objective & Engineering Purpose
1. Source Initial Ingestion	`Csv.Document(File.Contents(...))`	**Objective:** Connect to the raw local CSV file (`Metro_zhvi_uc_sfrcondo_tier_0.33_0.67_sm_sa_month.`) and ingest raw rows into the Power Query staging engine.
2. Changed Type Initial Typing	`Table.TransformColumnTypes(...)`	**Objective:** Perform preliminary text parsing of imported CSV columns prior to promoting the header row.
3. Promoted Headers Header Promotion	Use first row as headers `Table.PromoteHeaders(...)`	**Objective:** Elevate Row 1 into official table column headers (`RegionI` `SizeRank`, `RegionName`, `RegionType`, `StateName`, and monthly da strings).
4. Changed Type Static Typing	Explicit datatype assignments on the 5 fixed columns	**Objective:** Lock `RegionID` and `SizeRank` as **Whole Numbers (123)**, lock `RegionName`, `RegionType`, and `StateName` as **Text (ABC)**.

Phase II (B): Power Query ETL — Unpivoting & Final Filtering

ETL Steps 5–8

Applied Step Identifier	Transformation Operation	Technical Objective & Engineering Purpose
5. Unpivoted Other Columns Core Transformation	Select 5 base columns → *Unpivot Other Columns* (`Table.UnpivotOtherColumns`)	**Objective:** Resolve horizontal date sprawl by collapsing all monthly date columns into two vertical columns: `Attribute` (month-end date string) and `Value` (property value).
6. Renamed Columns Schema Cleanup	`Table.RenameColumns` (`Attribute` → `Date`, `Value` → `HomeValue`)	**Objective:** Replace generic Power Query output headers with semantic business identifiers: **`Date`** and **`HomeValue`**.
7. Changed Type1 Datatype Refining	Cast `Date` to **Date** & `HomeValue` to **Decimal Number (`1.2`)**	**Objective:** Enable continuous calendar time-intelligence on `Date` and configure high-precision floating-point arithmetic for `HomeValue`.
8. Filtered Rows Anomaly Purging	`Table.SelectRows(#"Changed Type1", each ([StateName] <> ""))`	**Objective:** Exclude records where `StateName` is empty, automatically purging the `United States` national aggregate row and retaining only valid MSAs.
Create Dax Measures Table Model Organization	*Home* → *Enter data* utility in Power Query (`Queries [2]`)	**Objective:** Create a dedicated, empty measure container table named **`Dax Measures`** to house all analytical DAX formulas cleanly.

Post-ETL Data Quality Profiling Verification: As verified in the Power Query column quality bars (Figure 2), all 7 resulting columns achieve **Valid: 100%**, **Error: 0%**, and **Empty: 0%**.

Final Cleaned Fact Table Schema (ZHVI)

Column Name	Data Type	Quality Status	Sample Record (Row 1 — New York MSA)
`RegionID` / `SizeRank`	Whole Number (123)	100% Valid | 0% Empty	`RegionID = 394913` | `SizeRank = 1`
`RegionName` / `RegionType` / `StateName`	Text (ABC)	100% Valid | 0% Empty	`New York, NY` | `msa` | `NY`
`Date` / `HomeValue`	Date / Decimal (`1.2`)	100% Valid | 0% Empty	`1/31/2000` | `220013.1236`

7 Columns — Structured Long Schema

0% Errors — Zero Execution Faults

0% Empty — Zero Missing Values

8 Steps — Automated ETL Pipeline

Phase III (A): Corrected Star-Schema Data Model (Model View)

Corrected Model View

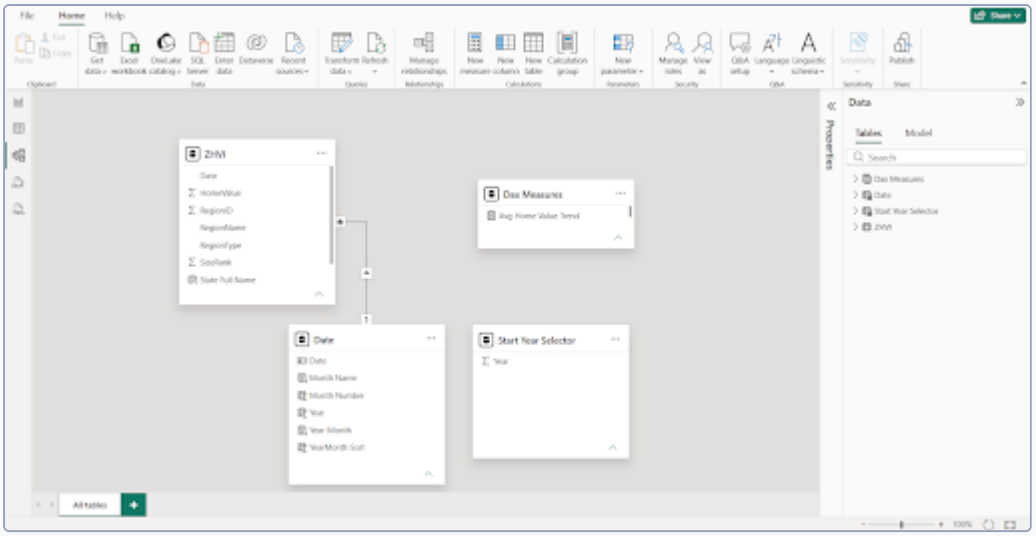


Figure 3: Corrected Star-Schema data model in Power BI Model View displaying all 4 tables and the active 1-to-Many (1:*) relationship

Table / Model Entity	Visible Fields & Attributes	Architectural Role & Relationship Behavior
1. ZHVI Fact Table (*)	`Date`, ∑ `HomeValue`, ∑ `RegionID`, `RegionName`, `RegionType`, ∑ `SizeRank`, `State Full Name`	Core analytical fact table on the Many (*) side of the relationship, enriched with calculated column `State Full Name` for geographic reporting.
2. Date Calendar Dimension (1)	`Date` (Key), `Month Name`, `Month Number`, `Year`, `Year-Month`, `YearMonth Sort`	Dedicated calendar dimension on the One (1) side, propagating single-direction (▲) time filters into `ZHVI[Date]`.
3. Start Year Selector Disconnected Table	∑ `Year` (Intentionally isolated without relationship lines)	Parameter table feeding the top `Start Year` slicer; queried dynamically via `SELECTEDVALUE('Start Year Selector'[Year])` without clipping trend axes.
4. Dax Measures Measure Container	Dedicated measure group displayed in the Data pane (`Tables / Model`)	Houses all analytical DAX measures in a centralized top-level folder so report authors can bind KPIs cleanly across all 4 pages.

Phase III (B): Calendar Dimension Inspection in Table View

Table View Analysis

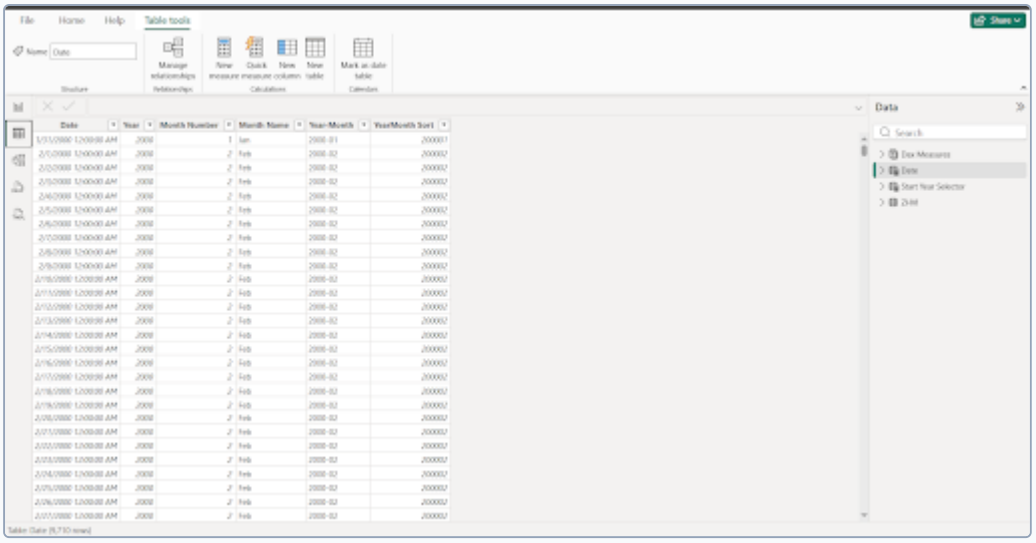


Figure 4: Power BI Table View inspection of the dedicated Date table (9,710 continuous daily rows starting 1/31/2000)

Calendar Column	Sample Values Shown	Technical Engineering Purpose in Power BI
`Date` Primary Key	`1/31/2000 12:00:00 AM`, `2/1/2000 12:00:00 AM`	Continuous daily timestamp key (**9,710 rows** total as shown in the bottom status bar) marked as the official Date Table key and linked `1:*` to `ZHVI[Date]`.
`Year`	2000 (Whole Number)	Extracts the 4-digit calendar year to drive the horizontal X-axis of the multi-year trend charts in *Overview* and *Geographic Analysis*.
`Month Number`	1, 2 (Whole Number)	Numeric month index (`1 = Jan`, `2 = Feb`, ..., `12 = Dec`) used as the sort-by column for `Month Name` so months sort chronologically rather than alphabetically.
`Month Name`	Jan, Feb (Text)	Abbreviatedthree-letter month label formatted for compact visual axes and tooltip drill-downs.
`Year-Month`	`2000-01`, `2000-02` (Text)	Concatenated `YYYY-MM` string enabling granular monthly time-series inspection across multi-year horizons.
`YearMonth Sort`	200001, 200002 (Integer)	Integer sorting key (`Year * 100 + MonthNumber`) ensuring chronological ordering of the `Year-Month` text column across all visuals.

Why a Continuous 9,710-Row Date Table Matters: DAX Time Intelligence functions and date-offset calculations (such as `EDATE(CurrentDate, -12)` and `DATEDIFF(..., DAY)`) require an unbroken calendar dimension to evaluate period-over-period comparisons accurately.

Phase III (C): DAX Measure Container & Dependency Architecture

Measure Inventory

Clean Measure Organization & Branching Strategy: All 10 analytical measures are housed inside the dedicated **`Dax Measures`** table at the top of the Power BI Data pane. Base valuation measures are branched into higher-order growth, compounding, and spatial ranking measures to guarantee single-source-of-truth accuracy.

#	DAX Measure Name	Format & Baseline Output	Upstream Measure Dependencies	Target Dashboard Page & Visual
01	`Home Value`	Currency ($115K–$300K)	Base Column: `ZHVI[HomeValue]`	*Overview* & *Geographic* Trend Charts
02	`Home Value (Start)`	Currency (Jan 31 Anchor)	`[Home Value]` + `Start Year Selector`	Internal Denominator for Growth & CAGR
03	`Latest Home Value`	Currency (**$300K**)	`[Home Value]` at `MAX(ZHVI[Date])`	*Investment Insights* Card & Scatter X-Axis
04	`Home Value (Average)`	Currency (**$192.79K**)	`[Home Value]` across Horizon	*Overview* Primary KPI Card (`AVG Home Value`)
05	**`CAGR %`**	Percentage (**3.80%**)	`[Home Value (Start)]`, `[Latest Home Value]`	*Investment Insights* Card & Top 5 Stable Table
06	`Cumulative Growth %`	Percentage (**169.82%**)	`[Home Value (Start)]`, `[Latest Home Value]`	*Overview* Card, Top/Bottom 10, Scatter Y-Axis
07	`YoY Growth %`	Percentage (**2.26%**)	`ZHVI[HomeValue]` + `EDATE(-12)`	*Overview* & *Investment Insights* KPI Cards
08	`Highest Growth Region`	Text (**`Kapaa, HI`**)	`[Cumulative Growth %]` + `TOPN(1, DESC)`	*Overview* Top Region KPI Card
09	`Lowest Growth Region`	Text (**`Camden, AR`**)	`[Cumulative Growth %]` + `TOPN(1, ASC)`	*Overview* Lowest Region KPI Card
10	`State Average Growth %`	Percentage (HI > 400%)	AVERAGEX over `[Cumulative Growth %]`	*Geographic Analysis* Bar Chart & Choropleth Map

Filter Context Synchronization: Every measure connected to `[Home Value (Start)]` automatically responds when a user changes the **`Start Year`** slicer (e.g., from 2000 to 2015 or 2020), dynamically recalculating Cumulative Growth, CAGR %, Top/Bottom 10 rankings, and State Averages in real time.

4 Price KPIs
Valuation Foundations

3 Growth KPIs
Total, CAGR & YoY %

2 Text KPIs
Top & Lowest Metros

1 Spatial KPI
Unweighted State Mean

Phase III (D): DAX Analytics — Foundation Price Measures (1 & 2)

DAX Measures 1–2

DAX Engineering Methodology: All analytical measures are authored using modular variable-driven syntax (`VAR ... RETURN`), optimizing VertiPaq evaluation and explicitly controlling filter context via `CALCULATE` and `REMOVEFILTERS`.

Measure (1): Baseline Average Home Value — `Home Value`

Output: Price Curve ($115K to $300K)

```
Home Value = AVERAGE('ZHVI'[HomeValue])
```

Calculation Logic	Computes the arithmetic mean of `ZHVI[HomeValue]` within the active visual filter context (by year, state, or metro region).
Visual Placement	Acts as the core foundation measure powering the **AVG Home Value Trend** area chart in *Overview* and the **Home Value and First State by Year** line chart in *Geographic Analysis*.

Measure (2): Initial Benchmark Value — `Home Value (Start)`

Output: Baseline Price at Jan 31 of Start Year

```
Home Value (Start) = VAR StartYear = SELECTEDVALUE('Start Year Selector'[Year]) VAR
StartDate = DATE(StartYear, 1, 31) RETURN CALCULATE( [Home Value], REMOVEFILTERS('Date'),
ZHVI[Date] = StartDate )
```

Calculation Logic	Captures the user-selected year from `'Start Year Selector'[Year]`, constructs the exact month-end anchor `DATE(StartYear, 1, 31)`, clears active calendar filters via `REMOVEFILTERS('Date')`, and evaluates `[Home Value]` at that baseline date.
Analytical Role	Serves as the initial investment denominator (V_{start}) for both `Cumulative Growth %` and `CAGR %` calculations.

Phase III (E): DAX Analytics — Latest Value & Period Average (3 & 4)

DAX Measures 3–4

Measure (3): Terminal Market Valuation — `Latest Home Value`

Dashboard Output: $300K

```
Latest Home Value = VAR LatestDate = CALCULATE( MAX('ZHVI'[Date]), REMOVEFILTERS('Date')
) RETURN CALCULATE( [Home Value], REMOVEFILTERS('Date'), 'Date'[Date] = LatestDate )
```

Calculation Logic	Finds the latest recorded timestamp in `ZHVI[Date]` ignoring date filters, then evaluates `[Home Value]` strictly at that final date (V_{end}).
Visual Placement	Drives the **Latest Home Value (300K)** KPI card in *Investment Insights* and forms the horizontal X-axis (entry price) of the regional Scatter Plot.

Measure (4): Multi-Year Horizon Mean — `Home Value (Average)`

Dashboard Output: $192.79K

```
Home Value (Average) = VAR StartYear = SELECTEDVALUE('Start Year Selector'[Year]) VAR
EndDate = CALCULATE( MAX(ZHVI[Date]), REMOVEFILTERS('Date') ) RETURN CALCULATE( [Home Value],
REMOVEFILTERS('Date'), ZHVI[Date] >= DATE(StartYear, 1, 31), ZHVI[Date] <= EndDate )
```

Calculation Logic	Averages property valuations across all monthly observations falling between `DATE(StartYear, 1, 31)` and `EndDate`.
Visual Placement	Powers the primary **AVG Home Value (192.79K)** KPI card on the *Overview* dashboard.

Key Analytical Distinction: Measure (3) captures current entry cost at the final month of the dataset (**$300K**), whereas Measure (4) computes the historical multi-year average across the entire 2000–2026 period (**$192.79K**).

Phase III (F): New Measure Deep-Dive — Compound Annual Growth (CAGR %)

New Measure Added (Measure 5)

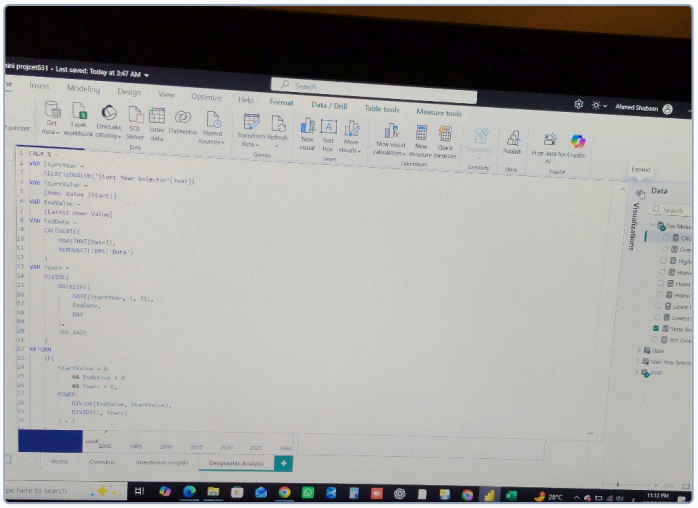


Figure 5: Live capture of the newly added CAGR % measure inside the Power BI Desktop DAX formula bar (lines 1–31)

Measure (5): Compound Annual Growth Rate — CAGR % **Output: 3.80% (Macro) | 7.31% (Kapaa, HI)**

```
CAGR % = VAR StartYear = SELECTEDVALUE('Start Year Selector'[Year]) VAR StartValue = [Home Value (Start)] VAR EndValue = [Latest Home Value] VAR EndDate = CALCULATE( MAX(ZHVI[Date]), REMOVEFILTERS('Date') ) VAR Years = DIVIDE( DATEDIFF( DATE(StartYear, 1, 31), EndDate, DAY ), 365.2425 ) RETURN IF( StartValue > 0 && EndValue > 0 && Years > 0, POWER( DIVIDE(EndValue, StartValue), DIVIDE(1, Years) ) - 1 )
```

Mathematical Formula	Implements the exact geometric compounding equation: **CAGR = $(V_{end} / V_{start})^{(1 / Years)} - 1$**, where exact elapsed years are computed by dividing total calendar days (`DATEDIFF(..., DAY)`) by **`365.2425`** (accounting for leap years).
Safety & Validation	Validates `StartValue > 0 && EndValue > 0 && Years > 0` prior to exponentiation, preventing division-by-zero or invalid power domain errors when filtering dynamic start years.
Dashboard Placement	Powers the **CAGR% (3.80%)** KPI card and ranks the **TOP 5 Most Stable Region** table on the *Investment Insights* page.

Phase III (G): DAX Analytics — Cumulative & YoY Growth (6 & 7)

DAX Measures 6–7

Measure (6): Total Horizon Appreciation — `Cumulative Growth %` — **Output: 169.82% (Macro) | 552% (Top Metro)**

```
Cumulative Growth % = VAR StartValue = [Home Value (Start)] VAR EndValue = [Latest Home Value] RETURN IF( StartValue > 0 && EndValue > 0, DIVIDE( EndValue - StartValue, StartValue ) )
```

Calculation Logic	Reuses `[Home Value (Start)]` and `[Latest Home Value]` via measure branching to compute total percentage appreciation: `(EndValue - StartValue) / StartValue`.
Visual Placement	Drives the **Total Growth (169.82%)** card, the **TOP 10 High-Growth** and **Bottom 10 Low-Growth** bar charts in *Overview*, and the Y-axis of the Scatter Plot in *Investment Insights*.

Measure (7): Trailing 12-Month Momentum — `YoY Growth %` — **Dashboard Output: 2.26%**

```
YoY Growth % = VAR CurrentDate = MAX('Date'[Date]) VAR CurrentValue = CALCULATE( AVERAGE(ZHVI[HomeValue]), REMOVEFILTERS('Date'), ZHVI[Date] = CurrentDate ) VAR PreviousYearValue = CALCULATE( AVERAGE(ZHVI[HomeValue]), REMOVEFILTERS('Date'), ZHVI[Date] = EDATE(CurrentDate, -12) ) RETURN IF( NOT ISBLANK(CurrentValue) && PreviousYearValue > 0, DIVIDE( CurrentValue - PreviousYearValue, PreviousYearValue ) )
```

Calculation Logic	Captures the latest month `CurrentDate` and shifts backward exactly 12 months via `EDATE(CurrentDate, -12)` to measure recent annual price velocity.
Visual Placement	Powers the **YOY Growth% (2.26%)** KPI card in both *Overview* and *Investment Insights*.

Phase III (H): DAX Analytics — Regional Extremes & State Averages (8–10)

DAX Measures 8–10

Measure (8): Top Outperforming Metro — `Highest Growth Region`

Text Output: Kapaa, HI

```
Highest Growth Region = VAR RegionTable = CALCULATETABLE(
ADDCOLUMNS(ALL(ZHVI[RegionName]), "@Growth", CALCULATE([Cumulative Growth %],
REMOVEFILTERS(ZHVI[StateName]))), REMOVEFILTERS(ZHVI[StateName]),
REMOVEFILTERS(ZHVI[RegionName]) ) VAR TopRegion = TOPN(1, FILTER(RegionTable, NOT
ISBLANK([@Growth])), [@Growth], DESC, ZHVI[RegionName], ASC) RETURN CONCATENATEX(TopRegion,
ZHVI[RegionName], ", ")
```

Logic & Placement	Builds a virtual table of all regions with their `[Cumulative Growth %]`, sorts descending (DESC) via `TOPN(1, ...)`, and displays **`Kapaa, HI`** in the *Overview* KPI card.

Measure (9): Slowest Growing Metro — `Lowest Growth Region`

Text Output: Camden, AR

```
Lowest Growth Region = VAR RegionTable = CALCULATETABLE(
ADDCOLUMNS(ALL(ZHVI[RegionName]), "@Growth", CALCULATE([Cumulative Growth %],
REMOVEFILTERS(ZHVI[StateName]))), REMOVEFILTERS(ZHVI[StateName]),
REMOVEFILTERS(ZHVI[RegionName]) ) VAR BottomRegion = TOPN(1, FILTER(RegionTable, NOT
ISBLANK([@Growth])), [@Growth], ASC, ZHVI[RegionName], ASC) RETURN CONCATENATEX(BottomRegion,
ZHVI[RegionName], ", ")
```

Logic & Placement	Identical virtual ranking table sorted ascending (ASC) to isolate the lowest appreciating region (**`Camden, AR`**) for the *Overview* KPI card.

Measure (10): Unweighted State Appreciation — `State Average Growth %`

Output: Hawaii Top State (>400%)

```
State Average Growth % = AVERAGEX( VALUES('ZHVI'[RegionID]), [Cumulative Growth %] )
```

Logic & Placement	Iterates over distinct `RegionID`s within each state via AVERAGEX so every metro carries equal weight; powers the bar chart and U.S. map in *Geographic Analysis*.



Figure 6: Executive Home screen displaying corporate Zillow branding, hero imagery, and interactive navigation buttons

Dashboard Element	Position & Component Type	Functional Role in User Experience (UX)
Zillow Logo & ZHVI Title	Top-Left Main Header	Establishes corporate visual identity, clearly indicating that the report analyzes official **Zillow Home Value Index (ZHVI)** residential data.
Overview Button	Interactive Blue Pill Button (Mid-Left)	Navigates users directly to Page 2 (*Overview*) via Power BI Page Navigation action to inspect macro trends and Top/Bottom 10 metros.
Investment Insights Button	Interactive Blue Pill Button (Mid-Right)	Navigates users to Page 3 (*Investment Insights*) for CAGR benchmarking, price-vs-growth scatter analysis, and the Top 5 Most Stable Regions table.
Geographic Analysis Button	Interactive Blue Pill Button (Center-Bottom)	Directs users to Page 4 (*Geographic Analysis*) featuring state-level rankings and the U.S. Choropleth heatmap.
Strategic Value Banner	Bottom-Left Callout Pill with Chart Icon	Communicates the core analytical mission of the dashboard: *"Determining an attractive and profitable pricing approach."*

Phase IV (B): Market Overview Dashboard (Overview Page)

Dashboard Page 2 of 4

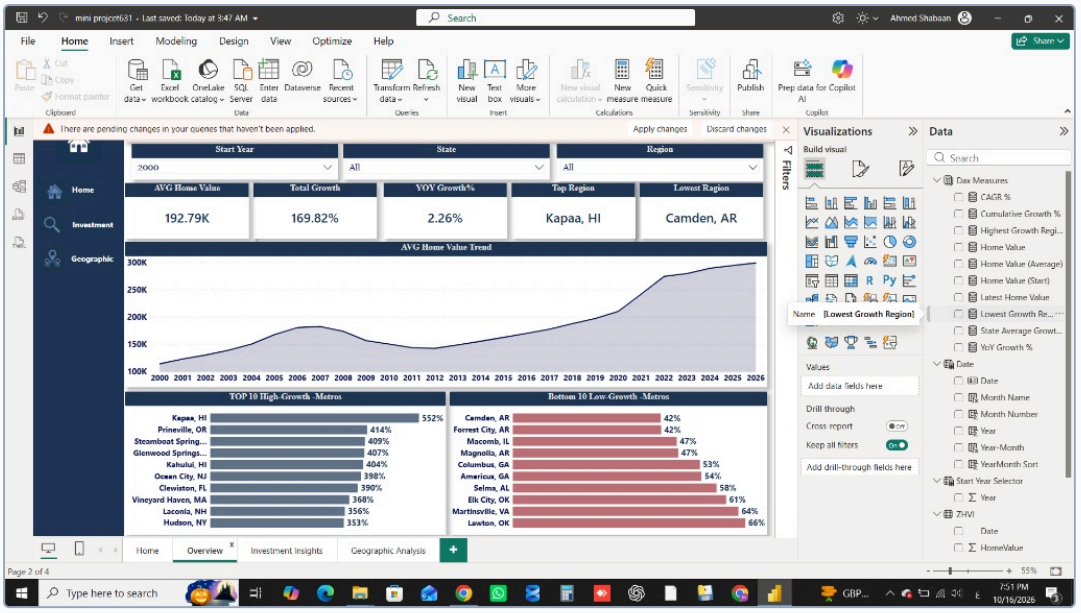


Figure 7: Overview Dashboard featuring top filter bar, 5 KPI cards, historical price trend curve, and Top/Bottom 10 metro rankings

Dashboard Component	Metric & Measure Binding	Analytical Interpretation (2000–2026 Baseline)
Top Slicer Bar	`Start Year: 2000` | `State: All` | `Region: All`	Sets the dynamic starting anchor year to 2000 while allowing instant filtering by any U.S. state or metropolitan area.
Primary KPI Cards (3)	`192.79K` | `169.82%` | `2.26%`	Displays multi-year average price (**$192.79K**), cumulative appreciation (**169.82%**), and trailing 12-month momentum (**2.26%**).
Top / Lowest Region Cards	`Kapaa, HI` | `Camden, AR`	Dynamically surfaces the #1 fastest-growing market (**Kapaa, HI**) and slowest-growing market (**Camden, AR**).
AVG Home Value Trend	`Home Value` by `Date[Year]` (2000–2026)	Captures the pre-2007 housing boom (~$185K), the 2008–2012 market correction (~$145K), and the post-2020 surge reaching **$300K** in 2026.
TOP 10 High-Growth Metros	`Cumulative Growth %` (Descending)	Led by `Kapaa, HI` (**552%**), `Prineville, OR` (**414%**), `Steamboat Springs, CO` (**409%**), `Glenwood Springs, CO` (**407%**), and `Kahului, HI` (**404%**).
Bottom 10 Low-Growth Metros	`Cumulative Growth %` (Ascending)	Identifies lagging markets: `Camden, AR` (**42%**), `Forrest City, AR` (**42%**), `Macomb, IL` (**47%**), and `Magnolia, AR` (**47%**).

Phase IV (C): Investment Insights Dashboard

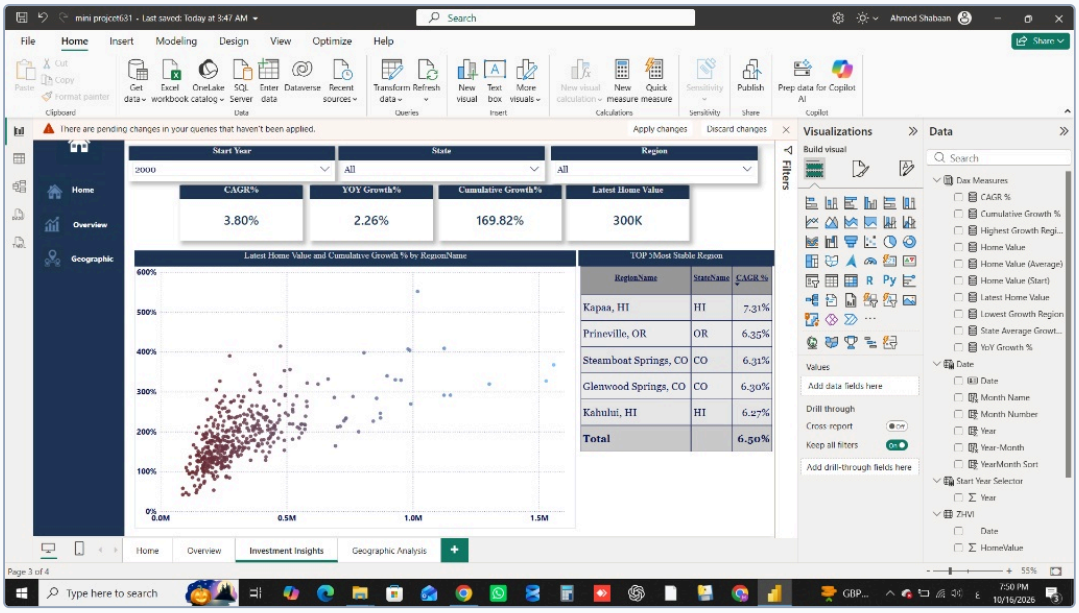


Figure 8: Investment Insights Dashboard highlighting CAGR%, Latest Home Value, Scatter Plot Matrix, and Top 5 Most Stable Regions

Dashboard Component	Metric & Measure Binding	Investment Application & Decision Value
Top KPI Strip (4 Cards)	`CAGR%: 3.80% | YOY: 2.26% | Cumul: 169.82% | Latest: 300K`	Pairs long-term compounding rate (**3.80%**) and total return (**169.82%**) directly alongside current market entry price (**$300K**) and recent momentum (**2.26%**).
Scatter Plot Matrix Value Discovery	X: `Latest Home Value` Y: `Cumulative Growth %` Detail: `RegionName`	Maps Growth vs. Entry Price across all U.S. metros. Most markets cluster between **$150K–$450K** on the X-axis and **100%–350%** on the Y-axis, enabling investors to spot high-return markets without $1M+ capital requirements.
TOP 5 Most Stable Region Table CAGR Ranking	Columns: `RegionName`, `StateName`, `CAGR %` (Combined Total: **6.50%**)	Lists the top 5 compounding metros outperforming the 3.80% benchmark: 1. **Kapaa, HI** (HI) — **7.31%** | 2. **Prineville, OR** (OR) — **6.35%** 3. **Steamboat Springs, CO** (CO) — **6.31%** | 4. **Glenwood Springs, CO** (CO) — **6.30%** | 5. **Kahului, HI** (HI) — **6.27%**.

Phase IV (D): Geographic Analysis Dashboard

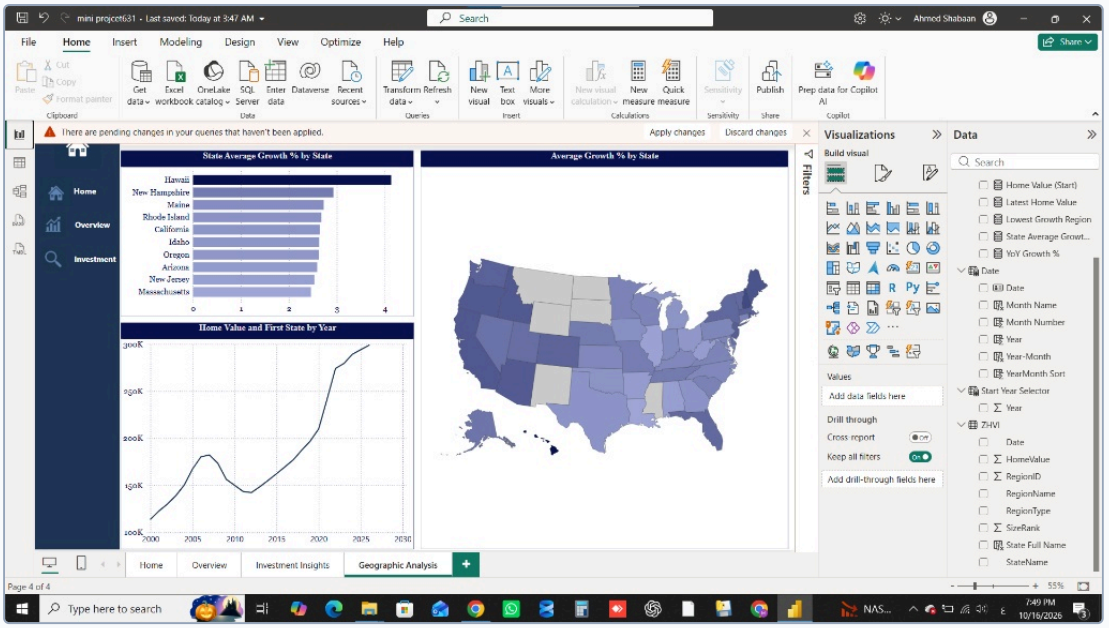


Figure 9: Geographic Analysis Dashboard featuring state growth bar chart, U.S. choropleth map, and temporal price curve

Dashboard Component	Associated Measure	Spatial & Economic Interpretation
State Growth Bar Chart *State Average Growth % by State*	`State Average Growth %` by `State Full Name`	Ranks U.S. states by unweighted regional appreciation: **Hawaii** ranks #1 (>4.0 / 400%), followed by **New Hampshire**, **Maine**, **Rhode Island**, **California**, **Idaho**, **Oregon**, **Arizona**, **New Jersey**, and **Massachusetts**.
U.S. Choropleth Heatmap *Average Growth % by State*	`StateName` & `State Average Growth %`	Visualizes spatial clustering across the United States; coastal West, Hawaii, Mountain West, and Northeast states display dark navy shading (high growth), whereas central states show lighter shading.
Temporal Price Curve *Home Value and First State by Year*	`Date[Year]` & `Home Value`	Traces the multi-year valuation trajectory from 2000 to 2026 ($115K to $300K) and dynamically cross-filters whenever a user clicks a specific state on the map or bar chart.
Dark Left Navigation Rail	Icons: `Home`, `Overview`, `Investment`	Provides a persistent left-hand navigation bar on all analytical pages for one-click switching between report modules.

Phase V (A): Project System Architecture & Binary File Structure

System Architecture

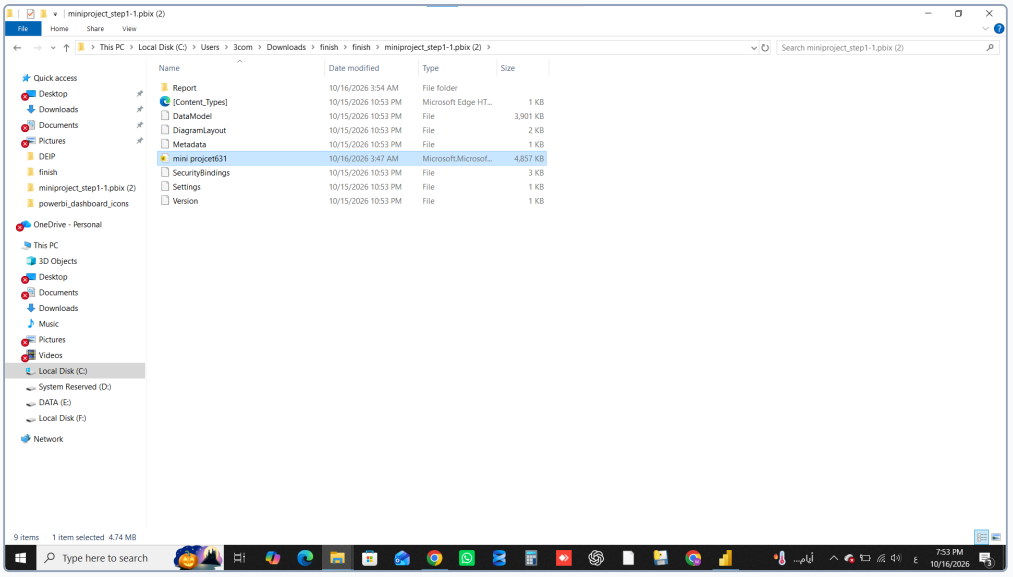


Figure 10: Windows File Explorer view showing the unpacked PBIX archive components and main executable file (mini projcet631)

File / Directory Name	Size & Type	Technical Role within Power BI Architecture
`mini projcet631`	4,857 KB (.pbix File)	Primary executable Power BI Desktop project bundle containing the entire data model, queries, DAX measures, and 4 interactive report pages.
`DataModel`	3,901 KB (Binary File)	Core VertiPaq in-memory columnar database storing the cleaned ZHVI rows, `Date` table (9,710 rows), relationships, and all 10 DAX measure definitions.
`Report`	File folder	Stores JSON visual layouts, custom Zillow color theme definitions, button navigation bindings, and canvas coordinates for all 4 pages.
`DiagramLayout`	2 KB (Binary File)	Persists the exact spatial coordinates and card properties of the 4 tables inside the Model View diagram canvas (Figure 3).
`Metadata` & `Settings`	1 KB each	Stores model metadata descriptors, Power Query M scripts, and report-level configuration preferences.
`SecurityBindings` & `Version`	3 KB & 1 KB	Enforces data protection bindings and records Power BI Desktop authoring version compatibility.

Phase V (B): Quality Assurance Audit & CSV Source Relinking Guide

QA & Maintenance

QA Audit Domain	Inspected Artifact	Verification & Validation Result
1. ETL Pipeline Integrity	Power Query ZHVI Table (Figure 2)	Verified all 8 Applied Steps execute in `1 m 41 s` with **100% Valid** quality and **0% Empty / Error** across all 7 columns; national aggregate row successfully removed.
2. Star-Schema & Calendar	Model View & Table View (Figures 3 & 4)	Confirmed active `1:*` single-direction link between `Date[Date]` (9,710 continuous daily rows) and `ZHVI[Date]`, alongside disconnected `Start Year Selector`.
3. DAX & Slicer Sync	`Dax Measures` Container & Formula Bar (Figure 5)	Cross-checked all 10 measures (including `CAGR % = 3.80%` and `Kapaa, HI = 7.31%`) across dynamic `Start Year`, `State`, and `Region` filter combinations.
4. UI/UX & Navigation	4 Report Pages (Figures 6–9)	Verified all interactive buttons on the Home portal and the dark left navigation sidebar across `Overview`, `Investment Insights`, and `Geographic Analysis`.

Step-by-Step Guide to Relink the CSV Data Source on Any Workstation

Step	Action in Power BI Desktop	Technical Instruction
Step 1	Open Power Query Editor	Launch `mini projcet631.pbix` in Power BI Desktop and click **Transform data** on the *Home* ribbon.
Step 2	Locate the `Source` Step	Select the **`ZHVI`** query in the left pane, then click the gear icon (⚙️) next to **`Source`** under *Applied steps* (or open **Advanced editor**).
Step 3	Update Local CSV Path	Browse to your local copy of `Metro_zhvi_uc_sfrcondo_tier_0.33_0.67_sm_sa_month.csv`, click **OK**, and click **Close & Apply**.

Golden Rule for Repository Portability: Updating only the `Source` file path inside Power Query preserves all subsequent 7 transformation steps, the 1-to-Many Calendar relationship, and all 10 DAX measures automatically without requiring any manual formula edits.

20 Pages	9,710 Rows	10 Figures	100% Passed
Complete Technical Manual	Verified Calendar Table	Clean Full-Width Captures	Final QA Sign-Off